Python Loops

50 Solved Programs with Questions and Answers

25 programs using the for loop + 25 programs using the while loop

Included in this PDF

- Question for every program
- Complete Python solution
- Short explanation
- Expected output where useful
- Beginner to intermediate progression

Teaching note: Run each program, predict its output, and trace the loop variable to help students understand how each iteration works.

Part 1: 25 Solved Programs Using for Loop

For each problem, read the question first, try to predict the result, then study the solution and explanation.

1. Question: Print numbers from 1 to 10

Answer (Python code):

```
for i in range(1, 11):
    print(i)
```

Explanation: `range(1, 11)` generates integers from 1 through 10. The stop value 11 is excluded.

Expected output / note: 1 2 3 4 5 6 7 8 9 10

2. Question: Print numbers from 10 to 1

Answer (Python code):

```
for i in range(10, 0, -1):
    print(i)
```

Explanation: The step `-1` decreases the value by one each time.

Expected output / note: 10 9 8 7 6 5 4 3 2 1

3. Question: Print even numbers from 1 to 20

Answer (Python code):

```
for i in range(2, 21, 2):
    print(i)
```

Explanation: Starting at 2 and using a step of 2 generates even numbers up to 20.

Expected output / note: 2 4 6 8 10 12 14 16 18 20

4. Question: Print odd numbers from 1 to 20

Answer (Python code):

```
for i in range(1, 21, 2):
    print(i)
```

Explanation: Starting at 1 and using a step of 2 generates odd numbers up to 19.

Expected output / note: 1 3 5 7 9 11 13 15 17 19

5. Question: Print the multiplication table of 5

Answer (Python code):

```
for i in range(1, 11):
    print(5, "x", i, "=", 5 * i)
```

Explanation: The loop multiplies 5 by each number from 1 to 10.

6. Question: Find the sum of numbers from 1 to 10

Answer (Python code):

```
total = 0

for i in range(1, 11):
    total += i

print("Sum =", total)
```

Explanation: `total` accumulates each number. The final sum is 55.

Expected output / note: Sum = 55

7. Question: Print squares of numbers from 1 to 10

Answer (Python code):

```
for i in range(1, 11):
    print(i, "square =", i ** 2)
```

Explanation: `i ** 2` calculates the square of each number.

8. Question: Print cubes of numbers from 1 to 10

Answer (Python code):

```
for i in range(1, 11):
    print(i, "cube =", i ** 3)
```

Explanation: `i ** 3` calculates the cube of each number.

9. Question: Find the factorial of a number

Answer (Python code):

```
n = 5
fact = 1

for i in range(1, n + 1):
    fact *= i

print("Factorial =", fact)
```

Explanation: The factorial of 5 is 5 × 4 × 3 × 2 × 1.

Expected output / note: Factorial = 120

10. Question: Count the digits of a positive integer

Answer (Python code):

```
n = 12345
count = 0

for digit in str(n):
    count += 1

print("Digit count =", count)
```

Explanation: `str(n)` converts the integer to text so the loop can visit each digit character.

Expected output / note: Digit count = 5

11. Question: Find the sum of digits

Answer (Python code):

```
n = 1234
total = 0

for digit in str(n):
    total += int(digit)

print("Sum of digits =", total)
```

Explanation: Each character is converted back to an integer and added to `total`.

Expected output / note: Sum of digits = 10

12. Question: Reverse a positive integer using a string

Answer (Python code):

```
n = 1234
reverse = ""

for digit in str(n):
    reverse = digit + reverse

print("Reverse =", reverse)
```

Explanation: Each new digit is placed before the reversed string built so far.

Expected output / note: Reverse = 4321

13. Question: Check whether a number is a palindrome

Answer (Python code):

```
n = 121

if str(n) == str(n)[::-1]:
    print("Palindrome")
else:
    print("Not a palindrome")
```

Explanation: Compare the string with its reverse. This solution uses slicing for the reversal.

Expected output / note: Palindrome

14. Question: Print the Fibonacci series for 10 terms

Answer (Python code):

```
a = 0
b = 1

for i in range(10):
    print(a)
    a, b = b, a + b
```

Explanation: Each term is the sum of the previous two terms.

Expected output / note: 0 1 1 2 3 5 8 13 21 34

15. Question: Check whether a number is prime

Answer (Python code):

```
n = 7
prime = True

if n < 2:
    prime = False
else:
    for i in range(2, n):
        if n % i == 0:
            prime = False
            break

if prime:
    print("Prime")
else:
    print("Not prime")
```

Explanation: A prime number has exactly two positive divisors: 1 and itself. The loop stops early if a divisor is found.

Expected output / note: Prime

16. Question: Print numbers divisible by 3 from 1 to 50

Answer (Python code):

```
for i in range(1, 51):
    if i % 3 == 0:
        print(i)
```

Explanation: `i % 3 == 0` checks whether the number is divisible by 3.

17. Question: Find the sum of even numbers from 1 to 100

Answer (Python code):

```
total = 0

for i in range(2, 101, 2):
    total += i

print("Sum =", total)
```

Explanation: The loop visits each even number from 2 to 100.

Expected output / note: Sum = 2550

18. Question: Find the largest among 5 positive numbers

Answer (Python code):

```
largest = None

for i in range(5):
    n = int(input("Enter a positive number: "))

    if largest is None or n > largest:
        largest = n

print("Largest =", largest)
```

Explanation: Enter five positive integers. The first number initializes `largest`; later numbers replace it if larger.

19. Question: Print ASCII values of uppercase letters A to Z

Answer (Python code):

```
for i in range(65, 91):
    print(chr(i), "=", i)
```

Explanation: `chr()` converts each code from 65 to 90 into its uppercase ASCII character.

20. Question: Count vowels in a string

Answer (Python code):

```
text = "Hello Python"
count = 0

for ch in text.lower():
    if ch in "aeiou":
        count += 1

print("Vowels =", count)
```

Explanation: The string is converted to lowercase, then each character is checked against the vowels.

Expected output / note: Vowels = 3

21. Question: Print a square star pattern

Answer (Python code):

```
for i in range(4):
    for j in range(4):
        print("*", end=" ")
    print()
```

Explanation: The outer loop controls rows and the inner loop prints four stars per row.

Expected output / note: * * * * (each row; 4 rows)

22. Question: Print a right-angled triangle

Answer (Python code):

```
for i in range(1, 6):
    for j in range(i):
        print("*", end=" ")
    print()
```

Explanation: The number of stars increases by one on each row.

23. Question: *

Answer (Python code):

Explanation: /

Expected output / note:

24. Question: Print a number triangle

Answer (Python code):

```
for i in range(1, 6):
    for j in range(1, i + 1):
        print(j, end=" ")
    print()
```

Explanation: The inner loop prints values from 1 up to the current row number.

25. Question: Calculate a power without using **

Answer (Python code):

```
base = 2
exponent = 5
result = 1

for i in range(exponent):
    result *= base

print("Result =", result)
```

Explanation: The loop multiplies the base by itself five times.

Expected output / note: Result = 32

26. Question: Find common divisors of two positive integers

Answer (Python code):

```
a = 12
b = 18

for i in range(1, min(a, b) + 1):
    if a % i == 0 and b % i == 0:
        print(i)
```

Explanation: A common divisor divides both numbers with no remainder.

Expected output / note: 1 2 3 6

Part 2: 25 Solved Programs Using while Loop

For each problem, read the question first, try to predict the result, then study the solution and explanation.

1. Question: Print numbers from 1 to 10

Answer (Python code):

```
i = 1

while i <= 10:
    print(i)
    i += 1
```

Explanation: The counter starts at 1 and increases until the condition becomes false.

Expected output / note: 1 2 3 4 5 6 7 8 9 10

2. Question: Print numbers from 10 to 1

Answer (Python code):

```
i = 10

while i >= 1:
    print(i)
    i -= 1
```

Explanation: The counter decreases by one on each iteration.

Expected output / note: 10 9 8 7 6 5 4 3 2 1

3. Question: Print even numbers from 1 to 20

Answer (Python code):

```
i = 2

while i <= 20:
    print(i)
    i += 2
```

Explanation: Incrementing by two visits only even numbers.

4. Question: Print odd numbers from 1 to 20

Answer (Python code):

```
i = 1

while i <= 20:
    print(i)
    i += 2
```

Explanation: Incrementing by two from one visits only odd numbers.

5. Question: Print the multiplication table of 7

Answer (Python code):

```
n = 7
i = 1

while i <= 10:
    print(n, "x", i, "=", n * i)
    i += 1
```

Explanation: The loop multiplies 7 by each multiplier from 1 to 10.

6. Question: Find the sum of numbers from 1 to 10

Answer (Python code):

```
i = 1
total = 0

while i <= 10:
    total += i
    i += 1

print("Sum =", total)
```

Explanation: The total is updated each iteration; the result is 55.

Expected output / note: Sum = 55

7. Question: Print squares from 1 to 10

Answer (Python code):

```
i = 1

while i <= 10:
    print(i ** 2)
    i += 1
```

Explanation: `i ** 2` calculates the square of the current number.

8. Question: Print cubes from 1 to 10

Answer (Python code):

```
i = 1

while i <= 10:
    print(i ** 3)
    i += 1
```

Explanation: `i ** 3` calculates the cube of the current number.

9. Question: Find the factorial of a number

Answer (Python code):

```
n = 5
fact = 1
i = 1

while i <= n:
    fact *= i
    i += 1

print("Factorial =", fact)
```

Explanation: The factorial of 5 is 5 × 4 × 3 × 2 × 1.

Expected output / note: Factorial = 120

10. Question: Count digits of a positive integer

Answer (Python code):

```
n = 12345
count = 0

while n > 0:
    count += 1
    n //= 10

print("Digit count =", count)
```

Explanation: Integer division by 10 removes the last digit. This example assumes n is positive.

Expected output / note: Digit count = 5

11. Question: Find the sum of digits

Answer (Python code):

```
n = 1234
total = 0

while n > 0:
    digit = n % 10
    total += digit
    n //= 10

print("Sum =", total)
```

Explanation: `% 10` extracts the last digit and `// 10` removes it.

Expected output / note: Sum = 10

12. Question: Reverse a positive integer

Answer (Python code):

```
n = 1234
reverse = 0

while n > 0:
    digit = n % 10
    reverse = reverse * 10 + digit
    n //= 10

print("Reverse =", reverse)
```

Explanation: The extracted digit is appended to the right side of the reversed number.

Expected output / note: Reverse = 4321

13. Question: Check whether a number is a palindrome

Answer (Python code):

```
n = 121
temp = n
reverse = 0

while temp > 0:
    digit = temp % 10
    reverse = reverse * 10 + digit
    temp //= 10

if n == reverse:
    print("Palindrome")
else:
    print("Not a palindrome")
```

Explanation: The program reverses the number using arithmetic and compares it with the original.

Expected output / note: Palindrome

14. Question: Print Fibonacci series for 10 terms

Answer (Python code):

```
a = 0
b = 1
i = 0

while i < 10:
    print(a)
    a, b = b, a + b
    i += 1
```

Explanation: The next term is calculated from the previous two terms.

Expected output / note: 0 1 1 2 3 5 8 13 21 34

15. Question: Check whether a number is prime

Answer (Python code):

```
n = 7
i = 2
prime = True

if n < 2:
    prime = False

while i < n and prime:
    if n % i == 0:
        prime = False
    i += 1

if prime:
    print("Prime")
else:
    print("Not prime")
```

Explanation: The loop tests divisors from 2 up to n - 1, stopping when a divisor is found.

Expected output / note: Prime

16. Question: Print numbers divisible by 3 from 1 to 50

Answer (Python code):

```
i = 1

while i <= 50:
    if i % 3 == 0:
        print(i)
    i += 1
```

Explanation: The condition prints only values whose remainder on division by 3 is zero.

17. Question: Find the sum of even numbers from 1 to 100

Answer (Python code):

```
i = 2
total = 0

while i <= 100:
    total += i
    i += 2

print("Sum =", total)
```

Explanation: The loop accumulates even values from 2 through 100.

Expected output / note: Sum = 2550

18. Question: Find the largest among 5 positive numbers

Answer (Python code):

```
i = 1
largest = None

while i <= 5:
    n = int(input("Enter a positive number: "))

    if largest is None or n > largest:
        largest = n

    i += 1

print("Largest =", largest)
```

Explanation: Enter five positive integers. `largest` starts as None so the first entered value initializes it.

19. Question: Print ASCII characters from A to Z

Answer (Python code):

```
i = 65

while i <= 90:
    print(chr(i), "=", i)
    i += 1
```

Explanation: `chr()` converts each code from 65 to 90 into its uppercase ASCII character.

20. Question: Count vowels in a string

Answer (Python code):

```
text = "Hello Python"
i = 0
count = 0

while i < len(text):
    if text[i].lower() in "aeiou":
        count += 1
    i += 1

print("Vowels =", count)
```

Explanation: The index moves through each character and checks whether it is a vowel.

Expected output / note: Vowels = 3

21. Question: Print a square star pattern

Answer (Python code):

```
i = 1

while i <= 4:
    j = 1
    while j <= 4:
        print("*", end=" ")
        j += 1
    print()
    i += 1
```

Explanation: The outer loop controls four rows; the inner loop prints four stars per row.

22. Question: Print a right-angled triangle

Answer (Python code):

```
i = 1

while i <= 5:
    j = 1
    while j <= i:
        print("*", end=" ")
        j += 1
    print()
    i += 1
```

Explanation: The inner loop prints as many stars as the current row number.

23. Question: Print a number triangle

Answer (Python code):

```
i = 1

while i <= 5:
    j = 1
    while j <= i:
        print(j, end=" ")
        j += 1
    print()
    i += 1
```

Explanation: The inner loop prints numbers from 1 up to the current row number.

24. Question: Calculate a power without using **

Answer (Python code):

```
base = 2
exponent = 5
result = 1
i = 1

while i <= exponent:
    result *= base
    i += 1

print("Result =", result)
```

Explanation: The loop multiplies the base by itself exponent times.

Expected output / note: Result = 32

25. Question: Find common divisors of two positive integers

Answer (Python code):

```
a = 12
b = 18
i = 1

while i <= min(a, b):
    if a % i == 0 and b % i == 0:
        print(i)
    i += 1
```

Explanation: A number is printed only if it divides both input values without a remainder.

Expected output / note: 1 2 3 6

Important reminders

- In range(start, stop), the stop value is excluded.
- A while loop must update its state appropriately to avoid an unintended infinite loop.
- The examples that ask for positive integers assume positive input unless otherwise stated.
- Nested loops are used for row-and-column patterns.